Ex:-12:- What is remainder when 37^{273} devided by 7?

Ans:- $\frac{37^{273}}{7} = \frac{(5\times7+2)^{273}}{7}$ $\Rightarrow$ Rem $= 2^{273} > 7$ then continue

$\frac{2^{273}}{7} = \frac{(2^3)^{91}}{7} = \frac{8^{91}}{7} = \frac{(7+1)^{91}}{7} \Rightarrow R = 1$

(iii) When $\frac{f(x+a_1)+f(x+a_2)+f(x+a_3)+\cdots}{f(x)}$ then Rem $\rightarrow f(a_1)+f(a_2)+\cdots$

(iv) When $\frac{f(x-a_1)+f(x-a_2)+\cdots}{f(x)} \Rightarrow$ then Rem is $\Rightarrow f(-a_1)+f(-a_2)+\cdots$

Ex:-13:- What is remainder of $\frac{36\times37\times38\times39\times40}{35}$?

Ans:- $\frac{(35+1)(35+2)(35+3)(35+4)(35+5)}{35} \Rightarrow R \Rightarrow 1\times2\times3\times4\times5 > 35$ then continue.

$\frac{1\times2\times3\times4\times5}{35} = \frac{120}{35} \Rightarrow R = \boxed{15}$

↳ division formula:- divisor) dividend (Quotient

$\boxed{\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}}$ Remainder

Ex:-14:- When a number devided by 16 gives remainder 7, then what is the remainder when the same number devided by 4?

Required Number $= x = 16Q + 7$

Rem when devided by 4 $\Rightarrow \frac{16Q+7}{4} = \frac{16Q}{4} + \frac{7}{4} = 0 + \boxed{3}$

When 1st divisor (or) is multiple of 2nd divisor then Answer for this kind of questions is $\frac{1^{st}\text{ Rem}}{2^{nd}\text{ divi}}$ Remainder

$R \rightarrow \frac{7}{4} \rightarrow \boxed{R = 3}$

(13) LCM & HCF:-

↳ LCM:- Least Common Multiple of 2 (or) more given numbers.

↳ HCF (or) GCD:- Highest Common factor (or) Greatest Common divisor of 2 or) more given numbers. Prime

↳ LCM (or) HCF can be find in 2 ways (i) factorization (ii) Successive division methods.

Ex:- LCM & HCF of 7 & 12 ?

Ans:- LCM-1:-

2	7, 12
2	7, 6
3	7, 3
7	7, 1
	1, 1

$\Rightarrow$ LCM $= 2\times2\times3\times7 = \underline{84}$

LCM-2:- Factorization of Prime method:-

$7 = 7^1$, $12 = 2\times2\times3 = 2^2\times3^1$.

LCM = Product of all primes in the 2 numbers

$= 7^1\times2^2\times3^1 = \underline{84}$

(or)

$7\times1 = 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, \boxed{84}, 91, 98 \ldots$

$12\times1 = 12, 24, 36, 48, 60, 72, \boxed{84}, 96, 108, \ldots\ldots$

HCF-1:- Factorization:-

$7 = 1\times7 \rightarrow$ 2 factors 1 & 7

$12 = 2\times2\times3 = 2^2\times3^1 \Rightarrow$ 1, 2, 3, 4, 6 & 12 are factors

Highest common factor $\Rightarrow$ HCF $\rightarrow$ (1)

HCF-2:- Successive division method:-

7	12	1
	7	
	5	1
	7	
	5	2
	4	
	1	

1	2	2
	2	
	0	

which divisor gives Rem '0' that is HCF $\rightarrow$ [1]

(132) Notes:-

(i) LCM $(x,y) \times$ HCF $(x,y) = x \times y$

(ii) HCF of co-primes is '1'.

(iii) HCF of fractions $= \left(\frac{\text{HCF of numerators}}{\text{LCM of denominators}}\right)$

(iv) LCM of fractions $= \left(\frac{\text{LCM of numerators}}{\text{HCF of denominators}}\right)$

(v) HCF of a numbers always devides its LCM.

(vi) Largest number which devides x, y & z to leave Remainder 'R' in each case is.

HCF of $[(x-R), (y-R), (z-R)]$

(vii) Largest number which devides x, y & z to leave same 'Rem'.

is HCF of $[(x-y), (y-z), (z-x)]$.

(viii) Largest number which devides x, y & z to leave remainders a, b & c is

HCF of $[(x-a), (y-b), (z-c)]$

(ix) Least ~~Largest~~ number which when devided by x, y & z and leaves a Rem 'R' in each case is.

$[$LCM of $(x,y,z)] + R$.

(x) LCM of No.s which are in the form of Product of powered Prime factors is.

↳ Product of the highest power of all prime factors.

(xi) HCF of No.s in the form of product of powered Prime factors is the ~~least powers~~ product of least powers of all common prime factors.